Policy gradients & actor–critic

Learn actions from rewards; learn values to guide the policy.

Learn a policy

Expected return
Policy-gradient derivation

Guide its learning

Baseline and advantage
Actor and critic

30 min mathematics · 10 min visual demo · 5 min recap

Why tables become impractical

Continuous observations and actions make discretization expensive.

4 state dimensions

100 bins each → 100^4 states

1 action dimension

100 bins → 100 actions

$$100^4 \times 100 = 10^{10}\ \text{Q entries}$$

Function approximation also exists for value-based RL.

Learn the policy directly

Observation → action distribution, rather than a table lookup.

Week 8 · Q estimates

Observation → Q table
Read action values
Choose an action

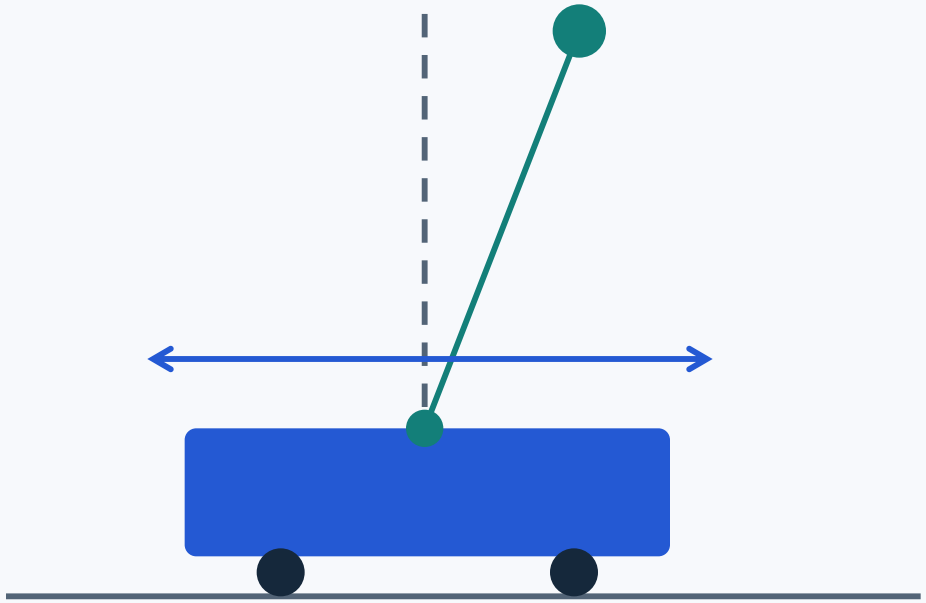
Week 9 · Actor

Observation → network
Describe likely actions
Sample an action

Reward feedback supplies learning signals; there are no expert action labels.

Our balancing task

Move the cart to keep the pole upright.



Observe

Position, angle
Two velocities

Act Continuous command

MuJoCo supplies dynamics and reward-bearing experience.

A stochastic continuous policy

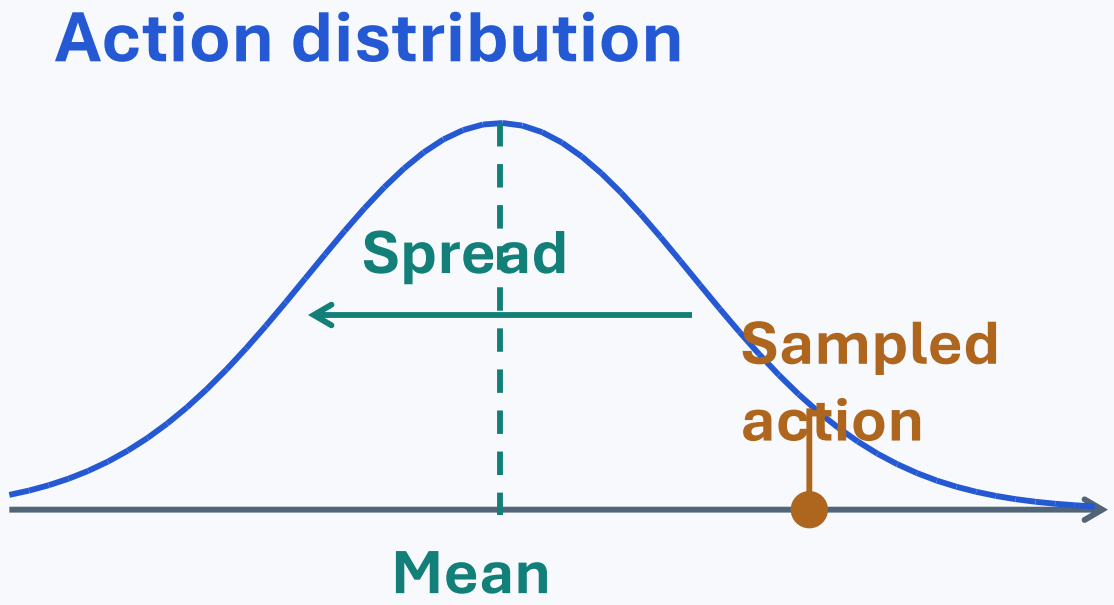
A distribution describes which actions are likely.

Observe

Robot state

Policy

Mean + spread



Mean: central action. Spread: randomness. Respect actuator limits.

What are we optimizing?

Increase expected discounted return.

$$J(\theta) = \mathbb{E}_{\tau \sim \pi\theta}\left[\sum_{t=0}^{T-1} \gamma^t R_t \right]$$

$$\theta \leftarrow \theta + \eta \nabla_\theta J(\theta)$$

θ: policy parameters · η: step size · γ: reward discount

Where the policy gradient comes from

Differentiate trajectory likelihood, rather than simulator physics.

$$p_\theta(\tau) = \rho_0(s_0) \prod_t \pi_\theta(a_t|s_t)\, P(s_{t+1}|s_t,a_t)$$

$$\nabla_\theta\, p_\theta(\tau) = p_\theta(\tau)\, \nabla_\theta \log p_\theta(\tau)$$

$$\nabla_\theta \log p_\theta(\tau) = \sum_t \nabla_\theta \log \pi_\theta(a_t|s_t)$$

The initial-state law and environment dynamics do not depend on θ.

Reward-to-go and the gradient estimate

An action receives credit for future rewards.

$$G_t = \sum_{k=t}^{T-1} \gamma^{k-t} R_k$$

$$\nabla_\theta J = \mathbb{E}[\, \sum_t \gamma^t G_t \nabla_\theta \log \pi_\theta(A_t|S_t) \,]$$

$$\hat{g} = \text{average of these weighted scores over trajectories}$$

Keep γ^t for the stated discounted start-state objective.

REINFORCE intuition

Return weights the direction of an action log-gradient.

$$\theta \leftarrow \theta + \eta\, \gamma^t\, G_t\, \nabla_\theta \log \pi_\theta(a_t|s_t)$$

Positive weight

Encourage sampled likelihood.

Negative weight

Discourage sampled likelihood.

A local likelihood update is not an expert action label.

Why plain returns are noisy

The same policy can produce very different trajectories.

Different starts

Some states are easier than others.

Sampled actions

Different choices change the trajectory.

Delayed effects

Many decisions contribute to return.

Compare outcomes against a useful reference level.

The baseline idea

Subtract expected performance without changing the mean gradient.

$$G_t \rightarrow G_t - b(S_t)$$

$$\mathbb{E}_{a\sim\pi\theta}[\, b(s)\, \nabla_\theta \log \pi_\theta(a|s)\,]$$

$$= b(s)\, \nabla_\theta \int \pi_\theta(a|s)\, da = b(s)\, \nabla_\theta\, 1 = 0$$

b(s) is action-independent and held fixed in the actor gradient.

Baseline worked example

Judge an outcome relative to what was expected.

$$\text{Centered weight} = \text{return} - \text{baseline}$$

Return 8; baseline 6

8 − 6 = +2

Return 4; baseline 6

4 − 6 = −2

Raw returns may both be positive; centered weights can differ in sign.

Advantage: better than expected

Compare taking this action with following the policy on average.

$V^{\pi}(s)$: expected return from s under π

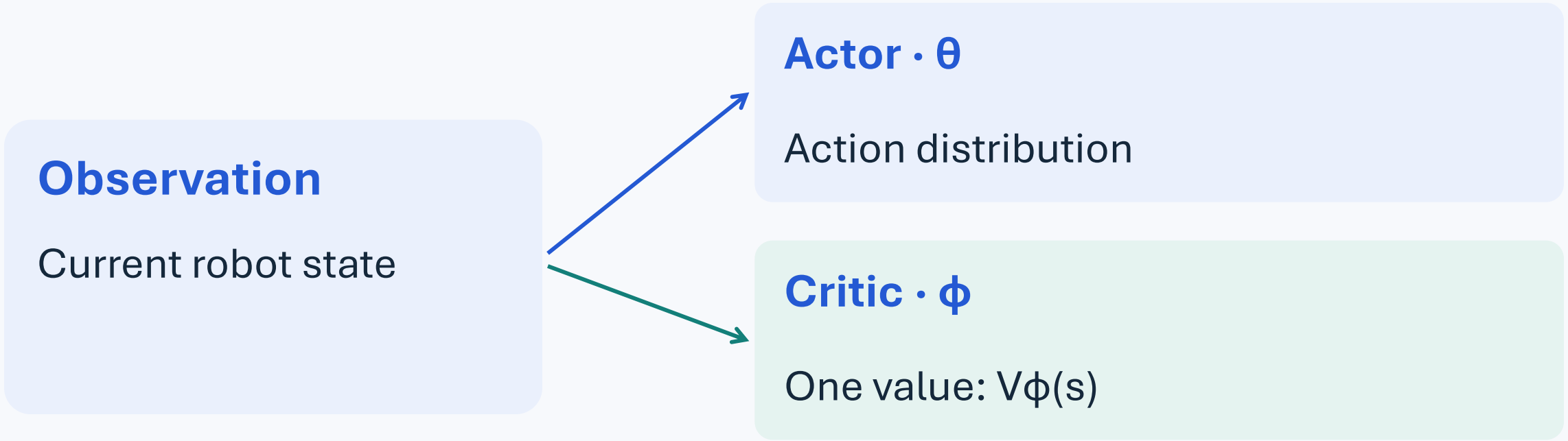
$Q^{\pi}(s,a)$: take a, then follow π

$$A^{\pi}(s,a) = Q^{\pi}(s,a) - V^{\pi}(s)$$

True advantage is unknown; learning uses an estimate Â.

Learn the baseline: introduce the critic

The actor chooses; the critic estimates expected return.



A critic is a value estimator, not an expert controller.

Actor–critic and the TD estimate

Compare a one-step return target with the current value prediction.

$$y_t = r_t + \gamma(1-d_t)\, V_\varphi(s_{t+1})$$

$$\hat{A}_t = y_t - V_\varphi(s_t)$$

$d_t = 1$: true termination; $d_t = 0$: continuation

Bootstrapping uses an estimate of future return.

Calculate one update signal

r = 1 · γ = 0.99 · V(s) = 5 · V(s') = 6

$$y = 1 + 0.99 \times 6 = 6.94$$

$$\hat{A} = 6.94 - 5 = +1.94$$

$$\text{Critic squared error} = (5 - 6.94)^2 = 3.7636$$

Encourage the sampled action; fit the critic toward 6.94.

Our five-step return target

Combine observed rewards with a learned continuation value.

$$y_t^{(n)} = \sum_{k=0}^{n-1} \gamma^k r_{t+k} + \gamma^n V_\varphi(s_{t+n})$$

$$n = 5\text{: } r_t + \gamma r_{t+1} + \cdots + \gamma^4 r_{t+4} + \gamma^5 V_\varphi(s_{t+5})$$

Actual boundary handling can shorten the return.

Termination: no bootstrap. Truncation: use the final observation.

Two losses, two learning roles

Treat advantage and return targets as fixed learning signals.

$$L_{actor} = -(1/N) \sum_t \log \pi_\theta(a_t|s_t)\ \text{stopgrad}(\hat{A}_t)$$

$$L_{critic} = (1/N) \sum_t [V_\varphi(s_t) - \text{stopgrad}(y_t)]^2$$

$$L = L_{actor} + 0.5\ L_{critic}$$

$$\theta \leftarrow \theta - \eta \nabla_\theta L_{actor} \quad | \quad \varphi \leftarrow \varphi - \eta \nabla_\varphi L_{critic}$$

Actor ascent becomes descent on a negative surrogate loss.

The A2C training loop

One update per fresh batch, then collect with the updated policy.



Collect

Fresh policy experience

→

Targets

Rewards + tail

→

Advantage

Target − value

→

Update

Both networks

8 environments × 5 steps = 40 transitions per update.

Exploration versus evaluation

Collect varied experience; test a frozen policy consistently.

Training

Sample from the policy
Learn from rewards
Update actor and critic

Evaluation

Use a representative action
Freeze all weights
Use held-out starts

Evaluation changes neither weights nor training environment streams.

PPO: discourage excessive policy changes

Keep the advantage signal; limit incentives for large likelihood changes.

$$\rho_t(\theta) = \pi_\theta(a_t|s_t) \,/\, \pi_{\text{old}}(a_t|s_t)$$

$$L^{\text{clip}} = \mathbb{E}[\, \min(\rho_t \hat{A}_t,\ \text{clip}(\rho_t, 1-\varepsilon_c, 1+\varepsilon_c)\, \hat{A}_t)\,]$$

$$\theta \leftarrow \theta + \eta \nabla_\theta L^{\text{clip}}$$

Overview only; our demo implements A2C.

SAC: reward plus entropy

Learn valuable actions while rewarding policy randomness.

$$L_\pi = \mathbb{E}_{s\sim D,\, a\sim\pi\theta}[\, \alpha_{ent} \log \pi_\theta(a|s) - \min_i Q_{\varphi i}(s,a) \,]$$

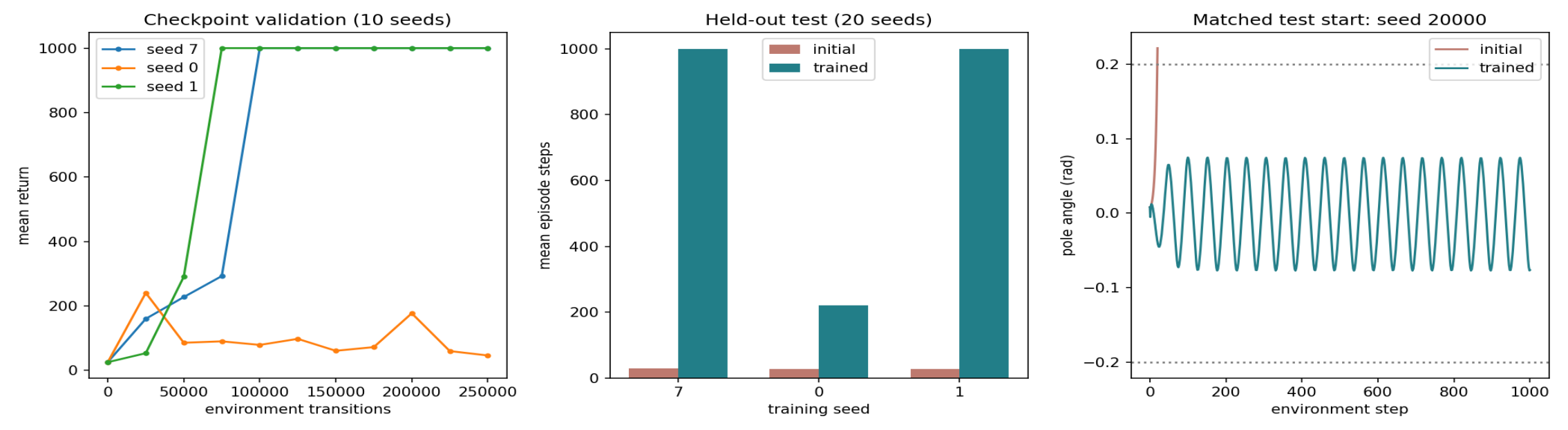
$$y = r + \gamma(1-d)[\, \min_i Q_{\varphi i}(s',a') - \alpha_{ent} \log \pi_\theta(a'|s') \,]$$

$$\theta \leftarrow \theta - \eta \nabla_\theta L_\pi;\ \text{critics fit } Q_{\varphi i}(s,a) \text{ to } y$$

Overview only; our demo implements A2C.

What the demonstration establishes

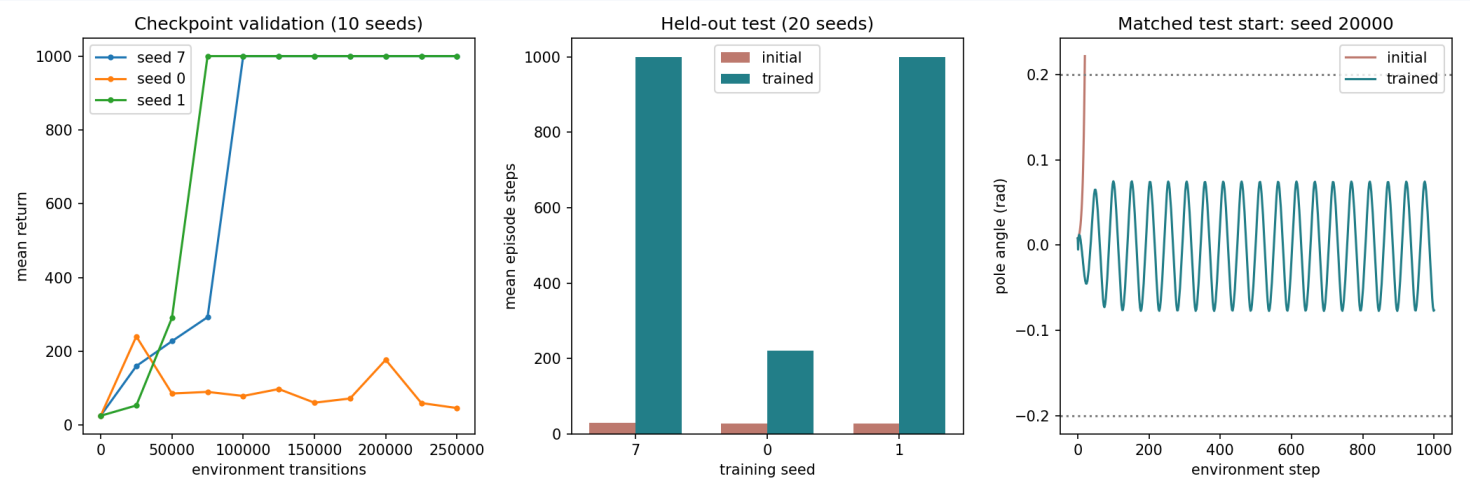
Matched comparisons show learning; multiple seeds show its variability.



Historical checkpoints: validation-selected, tested on separate starts.

Watch rewards become a balancing policy

Initial → trained on the same start → inspect → compare seeds.



Watch initial / trained.

Explain the two roles.

Compare three seeds.

2 min initial · 4 trained · 2 update signals · 2 evidence

Questions & recap

Explain the learning signal, the baseline, and the two roles.

Policy gradient

How do rewards change action likelihood?

Baseline

Why subtract a reference?

Actor–critic

Who acts, and who evaluates?

Policy gradient → baseline → advantage → actor–critic.

OPTIONAL / DEEPER DISCUSSION

Expanded trajectory-gradient derivation

Exact discounted start-state objective and practical rollout surrogate.

$$F(\tau) = \sum_k \gamma^k R_k; \quad J = \int p_\theta(\tau)\, F(\tau)\, d\tau$$

$$\nabla_\theta J = \int p_\theta(\tau)\, F(\tau)\, \nabla_\theta \log p_\theta(\tau)\, d\tau$$

$$= \mathbb{E}\left[\sum_t \gamma^t G_t \nabla_\theta \log \pi_\theta(A_t|S_t) \right]$$

$$\text{Demo surrogate: } -\text{mean}[\log \pi_\theta(a_t|s_t)\ \text{stopgrad}(\hat{A}_t)]$$

Uniform rollout weighting differs from the explicit γ^t estimator.

OPTIONAL / DEEPER DISCUSSION

Baseline cancellation & Gaussian gradient

The baseline centers the signal; the score sets its direction.

$$\mathbb{E}_a[b(s)\nabla_\theta \log \pi_\theta(a|s)] = b(s)\nabla_\theta 1 = 0$$

For $z \sim \text{Normal}(\mu,\sigma^2)$: $\partial \log p(z)/\partial\mu = (z-\mu)/\sigma^2$

Example: $z = 1, \mu = 0, \sigma = 1, \hat{A} = 2$

$\eta = 0.1$: $\mu \leftarrow 0 + 0.1 \times 2 \times 1 = 0.2$

Illustrative scalar ascent; the real network uses RMSprop.

OPTIONAL / DEEPER DISCUSSION

Multi-step targets and episode boundaries

Continue within the episode; never bootstrap through a reset.

Three observed rewards, tail value 2:

$$y = 1 + 0.99 + 0.99^2 + 0.99^3 \times 2 = 4.910698$$

True failure: tail $= 0$

Time cap: tail $=$ V(final observation), not V(reset state)

Longer targets use more experience and rely less on the critic.

OPTIONAL / DEEPER DISCUSSION

Bounded density and gradient separation

Implementation reference: fixed samples and fixed return targets.

$$z \sim \mathrm{Normal}(\mu_\theta(s), \sigma^2); \quad a = 3\tanh(z)$$

$$\log \pi_\theta(a|s) = \log p_\theta(z|s) - \log[3(1-\tanh^2 z)]$$

Fixed action sample | **Fixed advantage** | **Fixed return target**

Specific action transformation belongs here, not on the introductory slide.

OPTIONAL / DEEPER DISCUSSION

A2C, PPO, and SAC in context

Different update mechanisms; shared actor–critic ideas.

A2C · this demo

Fresh rollouts
V critic
One batch update

PPO · overview

Rollout minibatches
V critic
Clipped objective

SAC · overview

Replay data
Twin Q critics
Entropy objective

Only A2C is implemented in this week's demo.